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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Dsaise
Project / Solution Name	HalaTuju
Name (Leader)	Leow Shen En
Email & Phone (Leader)	leowshenen@gmail.com, 018-2335269
University Affiliation(s)	Universiti Malaya
Live Prototype URL	https://halatuju.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/shenenleow10521/SdcHackathon (committee accounts added as collaborators)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A JavaScript browser app in English and Bahasa Melayu (My skills, Explore map, Plan, My goal, For HR) is served by Node 22, TypeScript and Express on a Replit Reserved VM. CVs, pasted text or a task quiz enter the API. Local Tesseract OCR reads scans, and a scrubber removes personal details before any AI call. A language model splits the CV into phrases. BAAI/bge-m3 on RunPod Serverless matches them against 13,939 ESCO skills, and a second model call confirms the skills and current job. Our ridge model, trained on a RunPod GPU, scores each skill's AI exposure. Fixed formulas in a shared engine set match scores, route time and cost, Slow/Steady/Fast scenarios, SDG contributions and 1,000-future stress tests. A scheduled researcher collects dated evidence from 74 trusted domains. Quotes are checked, and three voter agents on two vendors' models decide: 2–3 of 3 apply, 1 of 3 goes to a person. Every vote is logged and every change can be undone. Replit PostgreSQL 16 stores profiles, knowledge versions, votes and goals. Sign-in uses Replit-managed Clerk, with guest access. GitHub Actions runs the type-check and 543 tests.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node 22; TypeScript 7; Express 5.3	Server and HTTP API	MIT / Apache-2.0
zod 4.6; multer 2.5; unpdf 1.9	Validation, uploads, PDF extraction	MIT / Apache-2.0
tesseract.js 7; eng and msa models	Local OCR	Apache-2.0
pg 8.23; PGlite 0.5	PostgreSQL access and local tests	MIT / Apache-2.0
fast-xml-parser 5.11; dotenv 18	Feed parsing and local config	MIT / BSD-2-Clause

Name & Version	Purpose in Project	License
3d-force-graph 1.79; three.js; GSAP 3.12	3D map and animation	MIT / GreenSock standard
Lucide; Geist font	Icons and typography	ISC / SIL OFL 1.1
@clerk/express 2.1; @clerk/clerk-js 6.40; @clerk/ui 1.41; http-proxy-middleware 4.2	Sign-in and sessions (Replit-managed Clerk)	MIT
Vitest 5; supertest 7	543 automated tests	MIT
Python 3.11; PyTorch; sentence-transformers; scikit-learn; NumPy; pandas; SciPy	Model training and evaluation	Python / BSD / Apache-2.0 as applicable

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
KrackedDevs LLM proxy (OpenAI-compatible)	GPT-5.5, gpt-6-sol and DeepSeek-V4-Pro for CV reading, explanations, claim review and the three-agent vote	Sponsor-provided
Google Gemini API	Fallback models when the KrackedDevs key is not set; embedding fallback	Free tier
RunPod Serverless API (endpoint a7ipau1nmqogxs)	BAAI/bge-m3 embeddings	Sponsor credit
RunPod GPU Cloud API	Launch and monitor the final training pod	Sponsor credit
Tavily Search API	Search trusted domains for market signals	Free tier
ESCO REST API; data.gov.my / OpenDOSM API	Build-time taxonomy and wage data	Free
arXiv; OpenAlex; GDELT; Hacker News Algolia; RSS	Public research and news signals	Free
YouTube oEmbed	Build-time check that linked learning videos exist	Free
Clerk (via Replit)	Sign-in and sessions	Sponsor platform

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Reserved VM and PostgreSQL 16	Live app at halatuju.replit.app; its only database	Sponsor platform; Asia

Service & Provider	Purpose in Project	Plan / Region
RunPod GPU Cloud	Final ridge-model training on RTX 4090 pod 38jijj8fe5dazu	Sponsor credit; Secure Cloud
RunPod Serverless	bge-m3 runtime embeddings, endpoint a7ipau1nmqogxs	Sponsor credit; 0–1 workers
Supabase	PostgreSQL during local development only	Free tier
GitHub and Actions	Repository, type-check and test CI	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-5.5	OpenAI via KrackedDevs proxy	Pre-trained; main model (CV reading, explanations, researcher extraction); evidence-checker voter	Provider and sponsor terms
gpt-6-sol	via KrackedDevs proxy	Pre-trained; skill-matcher voter	Provider and sponsor terms
DeepSeek-V4-Pro	DeepSeek via KrackedDevs proxy	Pre-trained; reviewer model; sceptic voter	Provider and sponsor terms
gemini-3.5-flash-lite	Google Gemini API	Pre-trained; main model when the KrackedDevs key is not set	Gemini API terms
gemini-3.1-flash-lite	Google Gemini API	Pre-trained; reviewer model when the KrackedDevs key is not set	Gemini API terms
gemini-embedding-001	Google Gemini API	Pre-trained; fallback embeddings	Gemini API terms
BAAI/bge-m3	BAAI / Hugging Face; RunPod	Pre-trained; 1,024-d embeddings	MIT
HalaTuju ridge exposure model	Team; trained on RunPod	Self-trained on human task labels	Project Apache-2.0; labels MIT
Tesseract eng and msa LSTM	Tesseract OCR	Pre-trained; local OCR	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	**Source / URL**	**Type (public / private / self-collected / synthetic)**	**License**	**Personal Data? (Y/N)**
ESCO v1.2.1; ESCO–O*NET crosswalk	esco.ec.europa.eu	Public taxonomy / mapping	EC reuse with attribution	N
O*NET tasks and ratings	onetcenter.org	Public occupational data	CC BY 4.0	N
O*NET 30.2 Detailed, Intermediate and Generalized Work Activities	onetcenter.org	Public occupational data	CC BY 4.0	N
GPTs are GPTs task labels	github.com/openai/GPTs-are-GPTs	Public training / evaluation	MIT	N
ILO WP140; Anthropic Economic Index	ilo.org; huggingface.co/Anthropic/EconomicIndex	Public research	CC BY 4.0 / CC BY	N
DOSM / OpenDOSM; MyCOL 2024/25	data.gov.my; TalentCorp	Public labour data	CC BY 4.0 / facts cited	N
Public course listings and dated news / research	Providers; trusted websites	Public facts, links and short quotes	Source-specific; links cited	N after scrub
45 learning resources (links checked 30 Oct 2026)	YouTube, Microsoft Learn, NIST, GHG Protocol, IFRS Foundation, The IIA, ACCA, WBCSD/WRI, Wiley and others	Public links	Provider terms; linked only	N
27 AI-era skills with 1–3 public sources each	Public sources, cited per skill	Public facts, short quotes	Linked, cited	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
No-CV quiz tasks (4,017 O*NET tasks, chips worded by an LLM)	onetcenter.org	Public / derived	CC BY 4.0	N
50 CV benchmark cases; demo persona; sample catalogue; sample HR team	Created for project	Synthetic	Project Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic; Opus models)	Code, tests, debugging and documentation	Major
Claude agents (Anthropic)	Synthetic CVs, blind answer keys and judging	Moderate
Replit Agent (Replit)	Migrated sign-in to Replit-managed Clerk	Moderate
Google Gemini gemini-3.5-flash	A discarded first benchmark judge run	Minor
faster-whisper large-v3	Local transcription of organiser tutorials	Minor
OpenAI Codex (GPT-6)	Reviewing the project and the declarations	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team directed the design and checked AI-assisted code with a TypeScript type-check and 543 automated tests, including fairness and prompt-injection tests. Held-out synthetic CV evaluations, a blind privacy run and stored model metrics exposed weaknesses, which are reported in the repository. Source quotations are checked before use. Every agent vote is logged with its reason, and split votes go to a person. Route-changing updates can be undone. The team remains responsible for all submitted claims, licences and links.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod	US$25 GPU / Serverless credit	RTX 4090 training run of our AI-exposure model, and runtime bge-m3 embeddings; pod 38jijj8fe5dazu, endpoint a7ipau1nmqogxs
Replit	Core platform, Reserved VM and PostgreSQL	Live prototype at halatuju.replit.app; Replit PostgreSQL 16 holds the knowledge base; all keys in Replit Secrets
KrackedDevs	LLM API proxy	Runs every agent: CV reading, explanations, and the three-agent evidence vote on models from two vendors

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod provides the GPU training run and the matching embeddings; a local bge-m3 server, Gemini embeddings and labelled keyword matching are fallbacks. The KrackedDevs proxy runs the agents and gives the vote panel models from two vendors; Google Gemini can replace it through the same interface. Replit hosts the live app and database; the Node 22 and PostgreSQL app could run on a comparable host.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app stores confirmed skills and levels, current role, study hours, goals, weekly check-ins, skill proofs and saved plans under a profile ID. Sign-in uses Replit-managed Clerk; guests can upload a CV without an account, tied to a signed browser cookie. CV files are read in memory and not stored. Scans use local OCR; names, identifiers, contact details and addresses are scrubbed before CV text is sent to any AI model or to RunPod. Held-out synthetic tests found 99.4% removal after fixes, which is not a guarantee for real CVs. HR team views show a worker's progress only if they opt in. Users can delete their own profile data. Keys are kept in Replit Secrets; staff routes require role checks. Saved data is kept until the user deletes it; provider-side retention follows provider terms.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

A misread article could change advice. Trusted domains, checked quotes, a three-agent vote on two vendors' models, a person for split votes and undo reduce that risk. The vote replay has only one human-decided signal so far, so agent accuracy is not yet measured. Our exposure model has higher per-task error than a nearest-neighbour baseline, so exposure stays an estimate. Routes cover all 3,039 ESCO jobs, but cross-field bridges and AI-era skill links are partly machine-suggested and reviewed. The synthetic CV benchmark does not prove real-world outcomes; sample data is labelled. Provider quotas and RunPod cold starts can delay calls; fallbacks are labelled. CV parsing and personal-detail removal can fail on unfamiliar documents, so users review their skills.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

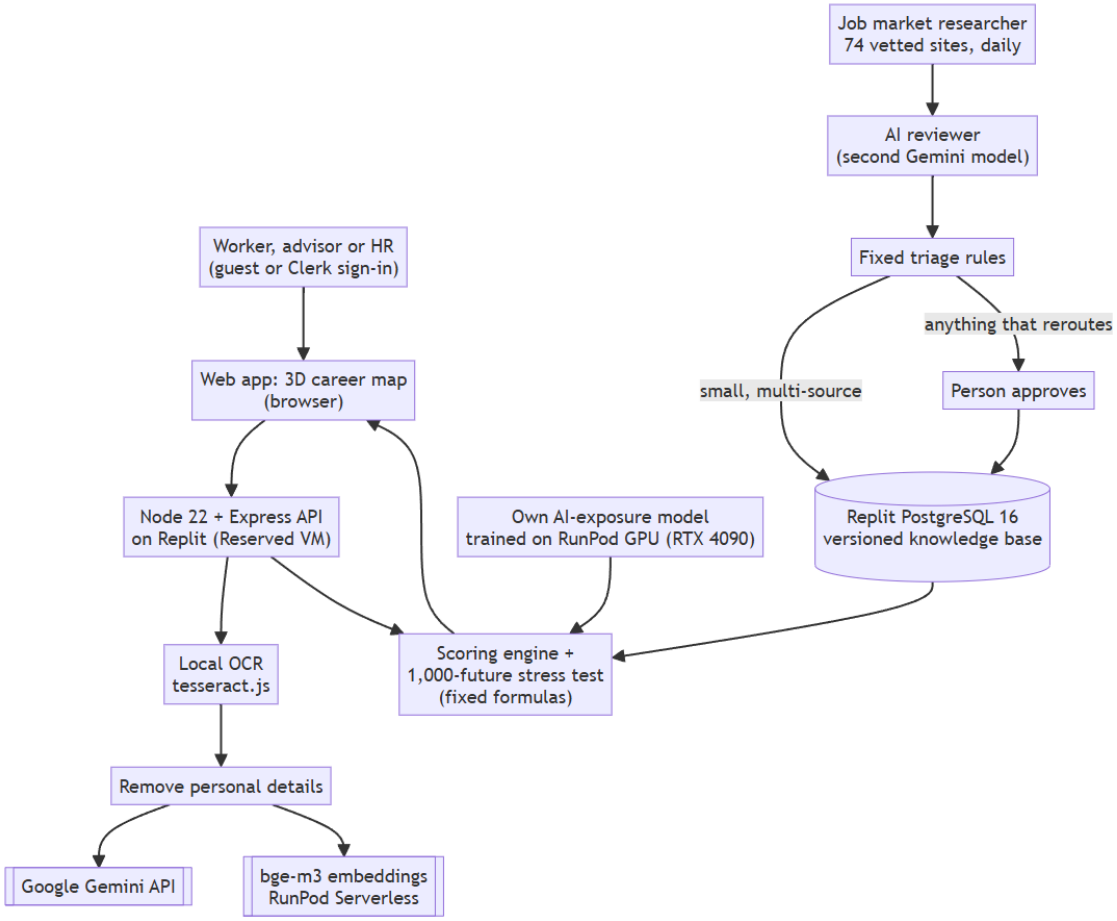
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



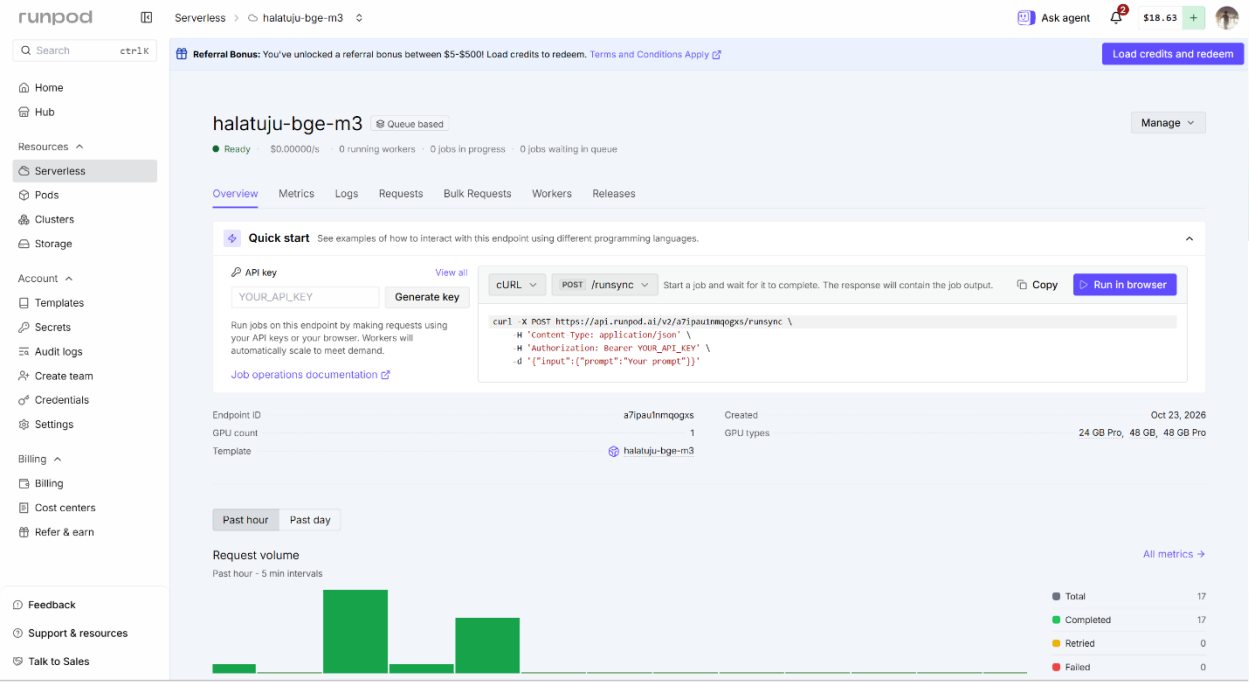
Date: 31 October 2026

Appendix A: System block diagram

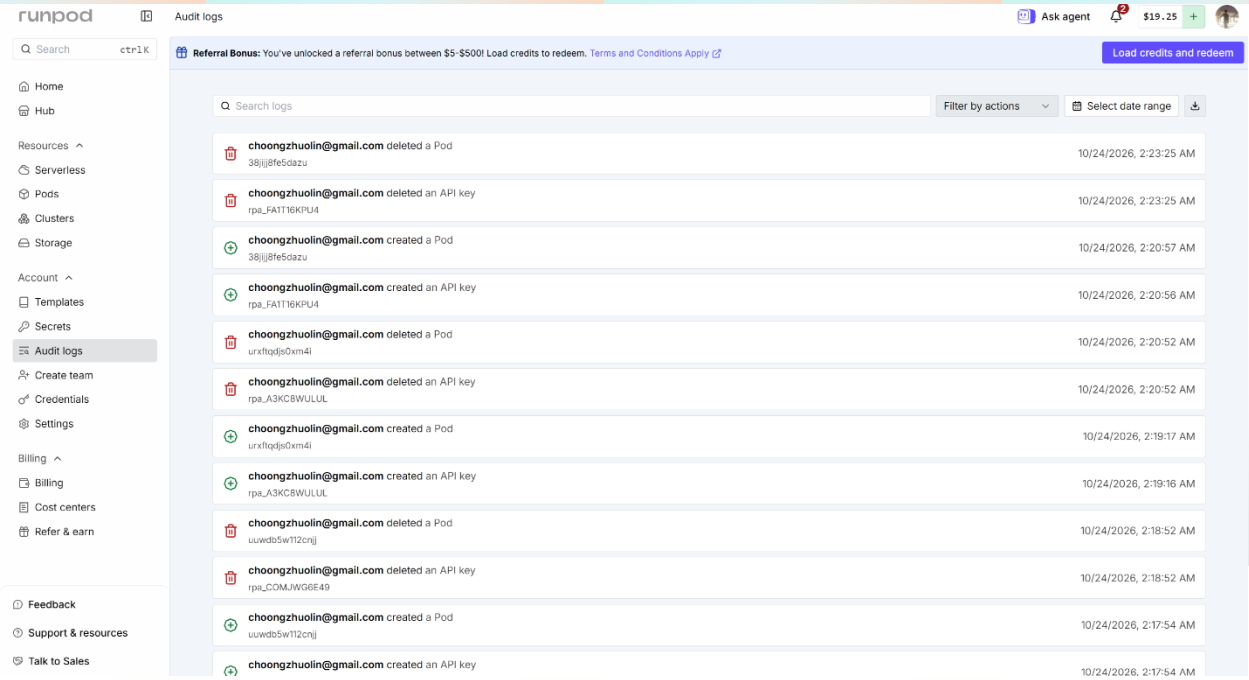


Appendix B: RunPod proof of use (team account)

Serverless endpoint halatuju-bge-m3, endpoint URL https://api.runpod.ai/v2/a7ipau1nmqogxs (ID a7ipau1nmqogxs); GPU types 24 GB Pro, 48 GB, 48 GB Pro; 0-1 workers. Used at runtime for bge-m3 embeddings.

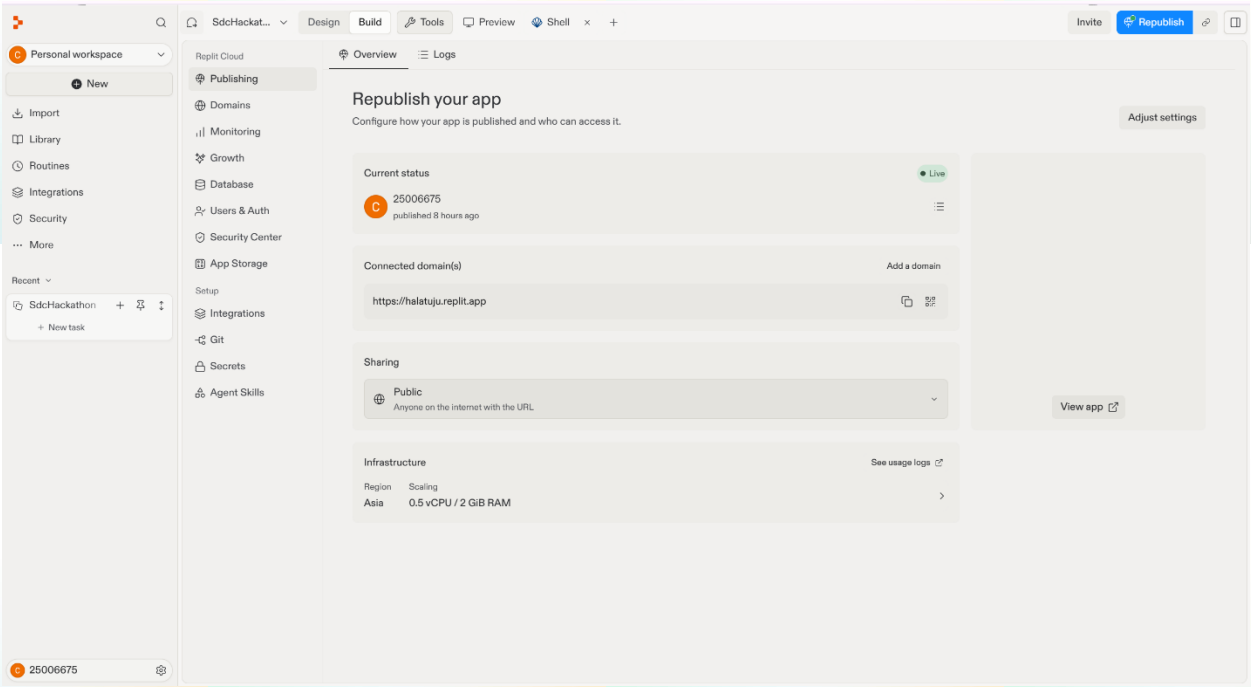


Training pod 38jijj8fe5dazu, GPU NVIDIA GeForce RTX 4090 (24 GB), Secure Cloud. Runtime: created 24 Oct 2026 02:20:57 MYT, deleted 02:23:25 MYT (about 2.5 minutes), as shown in the account audit log below. Run log: ml/artifacts/runpod_run.log in the repository.



Appendix C: Replit proof of use (team account)

Live deployment https://halatuju.replit.app from Replit account 25006675 (project SdcHackathon): Reserved VM, Asia region, status Live.



Replit Users & Auth (Clerk) in production: accounts created on the live app (user IDs only).

